Teamwork Agreement

For the MathWorks Classroom Challenge Projects

Project: Image-Based Defect Detection for Manufacturing Inspection

Team Name: Team 10

Instructor: EPP Team: Dr. Kevin Graham and Jiwon Kim

Mathworks Team: Dr. Harish Chinatakunta and Dr. Esperanza Guerrero

Program/Class: Engineering Pathway Program (EPP) 2026

Members and Roles

1. Marc Angarola
2. Molly Do
3. Basil Hamad
4. Alexis Arroyo

1) Purpose & Scope

Within MATLAB, design, test, and publish code for a virtual part defect inspection system that processes a specific set of images and will identify and classify the part type through a pretrained network, identify and produce evidence overlays and measurable metrics if defects are present, and deliver an AI classification decision based on the network's findings. A robustness and performance report based on repeated testing will be delivered.

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts:

2) Roles & Responsibilities

- **Marc:** Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved. Continuously organizes repository.
- **Marc:** Image Dataset Organizer: Will obtain and organize an image dataset to be used for processing based on defect classes and labels. Dataset will consist of 5 objects, each with a select few defect classes.
- **Molly and Basil:** Classical Image Processing Programmers: Implement classic image processing for an image dataset through preprocessing and segmentation for evidence overlay and masking while quantifying image defect evidence through metrics for sanity checking and debugging.
- **Marc:** AI Classification Programmer: Implements a pretrained network (or series of pretrained networks) that will be trained using transfer learning on the image dataset and outputs whether an image classifies as a pass or fail while also providing a confidence score for that classification.
- **Molly and Alexis:** Visualization and Evaluator Lead: Evaluates the inspection system along with the AI classifier using a test set of images from the dataset and documents and visualizes results of a confusion matrix/chart, yield and defect counts, and common failure cases of the classifier. Also evaluates and documents robustness using simulated variations of defect and defect-free images through changes in accuracy.
- **Everybody**: Quality Assurance, Validation, and Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions and is up to standard performance. Ensures data and code quality, and

tests/re-runs code to ensure robustness and reproducibility before moving on to the next task.
Note: All roles are responsible for documenting findings, results, and errors they encounter

3) Communication Norms

- Primary channel: **Discord**
- Secondary channel: **E-mail**
- Response time: **12 Hours on weekdays and 24 hours on weekends**
- Meeting cadence: **1 - 2 hours every Monday and Thursday in the late evening via Discord to share progress and establish next steps**
- Decision records: **All decision records will be recorded in the DECISION.md file in the repo.**

4) Collaboration & Tools

- Version control: https://github.com/QuantumBell
- Data management: **All datasets, parameters, and assumptions documented in:**

  image_data_set_info

File naming & style: **camelCase_ + descriptive names/variables.** (e.g. Snake_case, descriptive names)

- Backups: **Tag old versions by numbers, ex: V1.0, V.1.1, V1.2, a larger number refers to the newest revision of the file.**

5) Quality Standards (Definition of 'Done')

Program/system is reproducible, robust, and passes base cases + edge cases. All deliverables are met, no bugs or errors within the code, and all team members understand how each part of the program contributes to the next, what it does, and how all quantified data and results relates to the system. System yields a +90% success rate

6) Timeline & Milestones

- Week 1:
  - **Determine all team members' current weaknesses and strengths regarding current core background knowledge for the assigned project**
  - **Establish a weekly schedule for meeting time and roles for each member**
    - **Obtain background material through MathWorks on-ramping and other tutorials listed in the description of the assigned project for each member's assigned role, and explore, interpret, and connect the provided template to the background material**
  - **Begin working on organizing and implementing dataset**

- Week 2:
  - **Finish working on organizing and implementing dataset**
  - **Begin working on the classical image inspection function**

- Week 3:
  - **Continue and finish classical image evidence inspection function.**
  - **Begin working on training and integrating AI classifier with confidence score within its own function**
  - **Implement AI classifier into classical inspection function to finish image inspection system**
  - **Begin evaluating image processing and AI classification on a batch of images from the dataset**

- Week 4:
  - **Finish evaluating inspection system performance.**
  - **Begin and finish testing for robustness. Clean up and organize repository and MATLAB scripts**

7) Decision-Making & Conflict Resolution

- Default: **Group Consensus with PM facilitating**
- If blocked >24 hrs: **Time-limited debate, majority vote, PM documents decision**
- Technical disagreements: **Evidence-based comparison before vote**
- Escalation: **If unresolved, consult instructor/mentor with a concise memo.**

8) Workload, Accountability & Attendance

- Task board: **To-Do / Doing / Review / Done with section and due dates documented in README.md**
- Accountability: **Missed deadline → announce to team and follow through with recovery plan within 24 hrs; repeated misses trigger role redistribution**
- Attendance: **If you'll miss a meeting, give 24-hour notice and share updates**

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- **Establish image dataset to be used for inspection**
- **Load the dataset into the program and define defect classes and labels by organizing corresponding images**
    - **Choose 2 fail defects and 2 pass defects to be used for AI classification later**
- **Standardize and preprocess images, as well as segment images for evidence masking**
    - **Test and quantify the evidence to determine if the program is working as intended**

        - Explain any discrepancies, uncertainty, and/or error and document it
        - Document tested cases and their results and include an explanation
- Integrate AI classifier with confidence score within its own function
- Implement AI classifier into classical inspection function to finish image inspection system
    - Explain any discrepancies, uncertainty, and/or error and document it
- Evaluate system performance by placing report yield (rate of non-defected images detected) and defect detection rates on graphs and tables.
- Include uncertainty or error amongst different values for each rate. Include an explanation for all errors or uncertainty if extreme
    - Establish a confusion matrix to log results of AI classification for different defect classes of the test set
        - Document all data and formulate an explanation for results
- Ensure all code within scripts can be re-run, including regenerating necessary data such as plots, tables, and matrices for each run
- Perform multiple different robustness checks. Report and document accuracy and false-defect yield for each different check and compare and explain results for each
- Summarize documented evidence/data (include explanations of all aforementioned results above) of the project, interpreting the meaning (engineering application), limitations of the system, and next steps to improve and expand upon it
- Repository has clear structure, is organized with README's
- Peer review repo and project before submission

12) Agreement & Signatures

Name & Signature: **Marc Angarola** Date: 7/9/26

Name & Signature: **Alexis Arroyo** Date: 7/9/26

Name & Signature: **Molly Do** Date: 7/10/26

Name & Signature: **Basil Hamad** Date: 7/10/26